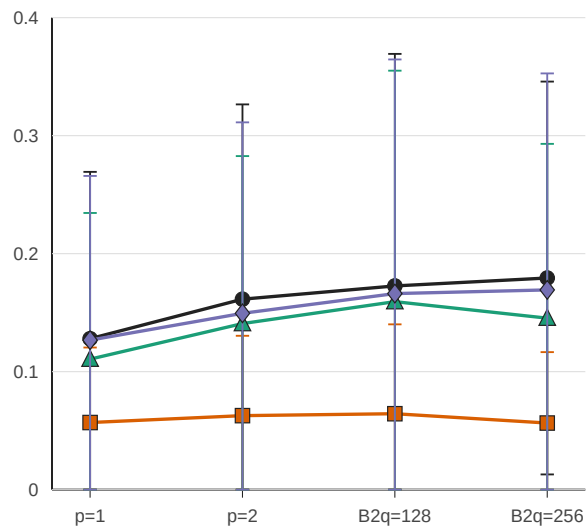


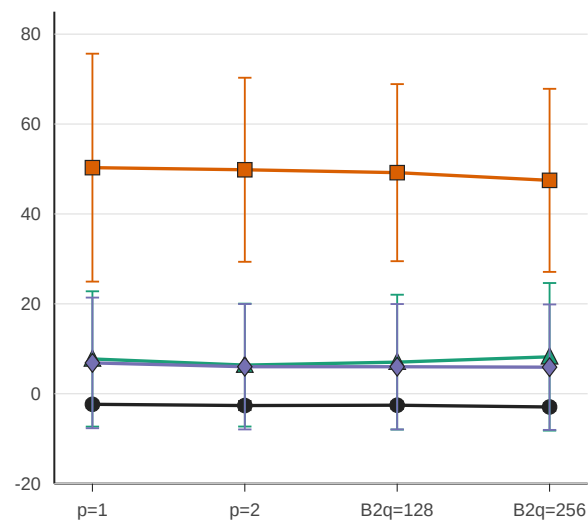
Secondary noiseless-QAOA diagnostics from the frozen runs

● Native ■ Full ▲ Selective ◆ Matched random

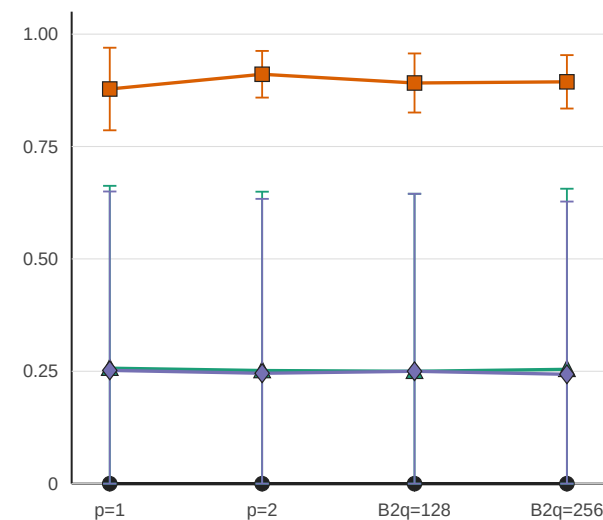
Cubic spin glass: Optimal-solution probability



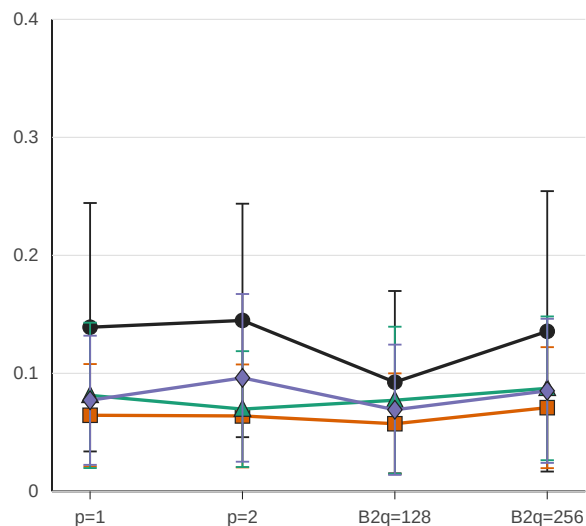
Cubic spin glass: Mean encoded energy



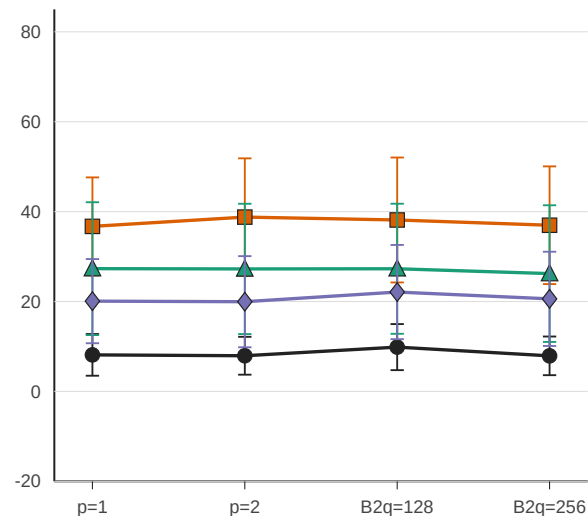
Cubic spin glass: Auxiliary inconsistency, R_inc



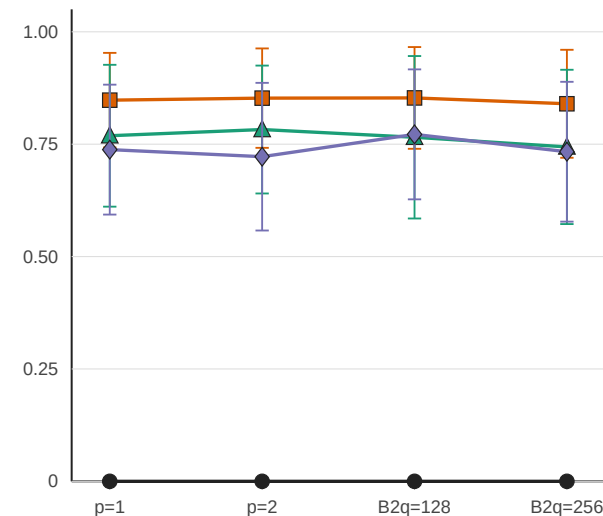
Max-3SAT: Optimal-solution probability



Max-3SAT: Mean encoded energy



Max-3SAT: Auxiliary inconsistency, R_inc



Points are means over seven raw instances; bars are instance-level 95% Student-t intervals. Restarts and then the five matched-random draws are averaged before the instance-level summary. Encoded energy is representation-specific and diagnostic only. B2q=128 includes stored p=0 gate-budget cases. Auxiliary inconsistency is the probability that at least one active relation $y_{ij} = x_i x_j$ is violated ($R_{inc} = 1 - P_{cons}$).